national Telecommunication Union (ITU) plays a vital role in administering and organising satellite services, allocating orbital slots, and combatting intentional interference. However, identifying and attributing cyber operations that cause harmful interference can be a complex task, given the intangible nature of such attacks.

In addition to international regulations, national frameworks for space safety play a significant role in protecting critical assets. Nations like the United States have established regulatory bodies and public-private partnerships dedicated to safeguarding satellites from cyber threats. The convergence of space and cyberspace requires continuous development and adaptation of national cybersecurity policies to ensure the resilience of satellite systems.

In the dynamic realm of large constellations, the Small-Sat market has witnessed significant growth. These smaller satellites, often weighing less than 180kg, offer advantages such as avoiding the Van Allen radiation belts, which reduce the risk of damage to their electronics. The market anticipates rapid expansion of large constellations with players like Starlink, OneWeb, Kuiper, Telesat, Hongyan, Hongyun, and Sphere leading the way.

As we explore the captivating world of large constellations, the importance of technical standards for safety and cybersecurity comes to the forefront. These standards form the backbone of secure and resilient operations within these vast networks of interconnected satellites. Ensuring adherence to these standards becomes crucial to maintaining the integrity and reliability of large constellations.

So, venture forth into this enthralling universe where cutting-edge technology, legal frameworks, and cybersecurity converge. Witness the cosmic dance of large constellations as they illuminate the skies, shaping the future of global connectivity while safeguarding their celestial pathways from cyber threats and ensuring the safe exploration of outer space.

Technical standards for cyber safety of large constellations

Verifying the source of cyber activities related to ground-satellite communication poses a significant challenge due to the complex nature of signal transmission. While attacks along the ground-satellite path are possible, the idea of an attack within the satellite-satellite segment remains hypothetical and unproven. Offensive satellites would require additional sensors and actuators not typically found onboard. Potential methods could involve using satellite awareness sensors to gather information about victim satellites or utilising third-party systems. Electromagnetic pulse actuators and radio frequency actuators might be employed to induce power system failures and GPS spoofing, respectively.

Advancements in satellite communications (SATCOM) and Global Navigation Satellite Systems (GNSS) have led to the development of a multilayer satellite system topology with a cross-layer design of multipath routing and a communication protocol stack. Utilising higher carrier frequencies, such as the X-through Ka-bands, offers advantages like reduced antenna aperture and higher data transfer rates. However, Ku/Ka bands are primarily suited for satellite-to-satellite communications for larger spacecraft.

To address congestion in lower RF frequencies, advanced programming techniques like the CCSDS low-density parity-check code (LDPC) family have proven effective in providing bandwidth and power trade-offs for CubeSat missions. Lasercom technologies, exemplified by the Optical Communications and Sensor Demonstration (OCSD) mission, have demonstrated successful data transfer in small spacecraft. Quantum key distribution (QKD) presents a potential solution for reducing cyber-attack risks by establishing private encryption keys between parties. The Zero-Trust Architecture, with its "never trust, always verify" approach, offers enhanced cybersecurity measures through threat protection and user access management.

Blockchain technologies have also shown promise in addressing security and trust issues in the space sector by providing decentralised and tamper-resistant data storage. However, limitations exist in terms of performance, scalability, privacy, and security vulnerabilities. An intrusion detection system (IDS) serves as the backbone of a cyber-resilient spacecraft, continuously monitoring telemetry and taking automatic actions through the intrusion prevention system (IPS) when threats are detected.

Emerging technologies are shaping standards for satellite-satellite communication and uplink/downlink operations in Small-Sats and CubeSats within large constellations. Implementing and adhering to these standards contribute to enhancing the safety of assets against hostile cyber operations.

Shared risks and shared responsibilities

Understanding cyber risks in the context of large satellite constellations is a challenging task. It is crucial to establish clear distinctions between the terms vulnerability, threat, and risk, as they represent different aspects of the problem. A vulnerability refers to a weakness